

Rain Liao

Adelaide, Australia

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[Linkedin](#) | [Project portfolio](#) | [Github](#)

PERSONAL SUMMARY

Data Engineer with practical experience designing and deploying scalable ETL pipelines, cloud infrastructure, and data storage solutions across AWS and Azure. Proficient in Python and SQL, with hands-on experience building containerized, cloud-deployed systems.

WORK EXPERIENCE

Aug 2025 - Oct 2025 Data science internship

University of Adelaide, SET Faculty (Adelaide Australia)

- Designed and deployed an end-to-end ETL pipeline (Python, Selenium, NLTK, Blob, Airflow) to ingest, clean, and structure 100,000+ unstructured social media posts into a query-ready format, applying statistical validation to topic modelling that identified 10 key subtopics within mental health discussions in Australia.
- Engineered a scalable data processing workflow supporting transformer-based modelling and a LangChain RAG system for downstream NLP analysis and insight extraction.

Aug 2024 - Mar 2025 Quantitative trading assistant

Australian Institute for Machine Learning (Adelaide Australia)

- Built and maintained a high-volume data pipeline processing 100GB+ of Binance market data, combining fine-tuned LLM sentiment signals with technical indicators for automated trading strategy execution using Python and Freqtrade.
- Containerized and deployed the trading bot to Azure VM, cutting environment setup time by 80% and delivering a reproducible production-ready pipeline for live trade execution.

Jul 2023 - Oct 2023 Special case technician

National Centre for Research on Earthquake Engineering (Taipei Taiwan)

- Scraped, processed, and structured large-scale weather and seismic datasets using Pandas and QGIS, storing results in AWS DynamoDB and S3 for scalable downstream access.
- Built and deployed a real-time data ingestion and alerting system on AWS (EC2, Lambda) integrating REST APIs and Google Maps API, with end-to-end monitoring via CloudWatch and SNS on a Linux-based infrastructure.

Jul 2025 - Present Front of House

Olive (Adelaide Australia)

- Provide excellent guest service through clear communication, teamwork, and quick problem-solving in a fast-paced fine dining environment.

PERSONAL PROJECT

2026

Real-Time Weather Analytics Pipeline

- Built a containerized end-to-end data pipeline using Docker Compose, integrating API ingestion, PostgreSQL, dbt transformations, and Airflow orchestration, with live analytics visualized on a [publicly accessible Superset dashboard](#).
- Publicly deployed from a home lab via Cloudflare Tunnel with no public IP or open ports, configuring the full networking stack including DNS, SSL/TLS, and traffic management for production-equivalent encrypted connectivity.

ACHIEVEMENTS

- Microsoft Azure Data Scientist Associate: DP-100
- Microsoft Power BI Data Analyst Associate: PL-300
- First-author conference publication – [ICRTCIS 2025 \(2026\)](#)
- Microsoft Fabric Data Engineer Associate: DP-700

TECHNICAL SKILLS

Languages: Python, T-SQL, R, C++, JavaScript
Data & Viz: ETL, DBT, APIs, Power BI, QGIS

Libraries: Pandas, Selenium, Scikit-learn, PyTorch, Airflow, PySpark
Cloud & DevOps: AWS, Azure, Cloudflare, Docker, Linux, GitHub

EDUCATION

MASTER OF DATA SCIENCE

University of Adelaide | Adelaide, SA
Feb 2024 – Dec 2025

BACHELOR of CIVIL ENGINEERING

Chung Yuan Christian University | Taoyuan, Taiwan
Sep 2018 - Jun 2023

LANGUAGE

English (IELTS 7.5)
Mandarin (Native)

REFERENCE

Available upon request